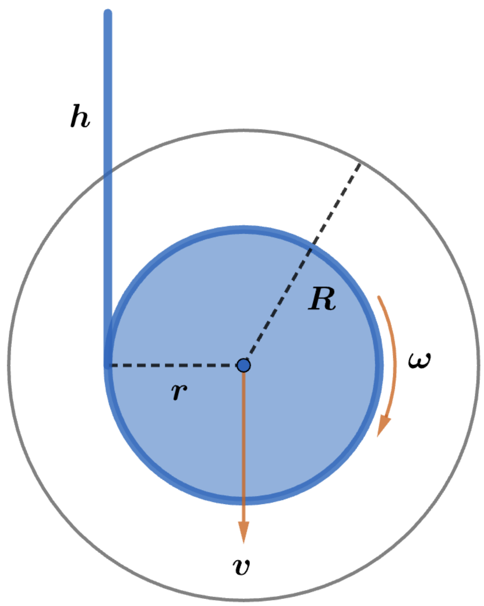
2024 F=ma Exam: Problem 25

Kevin S. Huang



We can find the yo-yo's velocity v at any point by conservation of energy. If it has fallen down by height h, then we have

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2$$

Since the yo-yo is made up of two stacked disks

$$I = \frac{1}{2}MR^2$$

The yo-yo moves down at the rate at which the string unwinds so

$$v = r\omega$$

Substituting these into the first equation,

$$Mgh = \frac{1}{2}Mv^2 + \frac{1}{2}\left(\frac{1}{2}MR^2\right)\frac{v^2}{r^2} = \frac{Mv^2}{2}\left(1 + \frac{R^2}{2r^2}\right)$$

Solving for v,

$$2gh = v^2\left(1 + \frac{R^2}{2r^2}\right)$$

$$v = \sqrt{\frac{2gh}{1 + \frac{R^2}{2r^2}}} = \sqrt{\frac{4ghr^2}{R^2 + 2r^2}}$$

It remains to write r in terms of h. Let the string have thickness w and total length L. By conservation of string,

$$\pi R^2 = Lw$$

$$\pi r^2 = (L - h)w$$

Thus,

$$v = \sqrt{\frac{4gh(L-h)}{L + 2(L-h)}} = 2\sqrt{\frac{gh(L-h)}{3L - 2h}}$$

In particular,

$$v(h = 0) = v(h = L) = 0$$

so there must have been downward and upward acceleration. Thus, the answer is $\boxed{\text{E}}$.